

LECTURE 6 – STATE ESTIMATION (II)

INTRODUCTION TO AUTONOMOUS SYSTEMS · MODULE III

Topics: Gaussian machinery · Closure theorems · Conjugacy · Kalman filter: predict, update, gain, algorithm

AGENDA

S	TOPIC	KEY IDEAS
3	Gaussian Machinery	Characteristic function · closure theorems · conjugacy
4	The Kalman Filter	Predict · update · gain · innovation · the algorithm

*Lecture 5 covered §1–§2: the hidden-state problem, the Bayes filter, and the corridor example.
§4 (cont.)–§7 — optimality, long-run behaviour, sensor fusion, particle filter, belief to policy — are in Lecture 7.*

RECAP – THE BAYES FILTER

Theorem (Bayes filter). Under the Markov dynamics and Markov observation assumptions:

$$\textbf{Predict: } \bar{b}_t(x_t) = \int p(x_t \mid x_{t-1}, a_{t-1}) b_{t-1}(x_{t-1}) dx_{t-1}$$

$$\textbf{Update: } b_t(x_t) = \eta p(o_t \mid x_t) \bar{b}_t(x_t), \quad \eta^{-1} = \int p(o_t \mid x) \bar{b}_t(x) dx$$

Predict spreads. Push the belief through the dynamics; process noise adds uncertainty.

Update sharpens. Reweight each state by how well it explains the observation.

Estimation alternates between these two stages.

LITERATURE

TOPIC	REFERENCE
Bayes, Gaussian and particle filters	Thrun, Burgard & Fox — <i>Probabilistic Robotics</i> (2005), Ch. 2–4
Rigorous filtering and smoothing	Särkkä — <i>Bayesian Filtering and Smoothing</i> (2013)
Kalman filter theory, Riccati, steady state	Anderson & Moore — <i>Optimal Filtering</i> (1979), Ch. 3–5
Original paper	Kalman — <i>A New Approach to Linear Filtering and Prediction Problems</i> (1960)

§3 · GAUSSIAN MACHINERY

PROVING THE CLOSURE PROPERTIES PROMISED IN LECTURE 4



**WHEN CAN THE PREDICT INTEGRAL AND THE UPDATE
PRODUCT BE COMPUTED EXACTLY?**



A FAMILY CLOSED UNDER BOTH STEPS

Answer: when the belief stays inside a parametric family that **both steps map back into itself**. Then the filter only tracks a few parameters instead of a whole function.

The Gaussian is such a family — **provided f and h are linear**.

THEOREM	KALMAN FILTER STEP
Linear transformation	Predict: $\bar{b}_t$ from b_{t-1}
Conditioning	Update: b_t from $\bar{b}_t$ and o_t
Gaussian–Gaussian conjugacy	Update (information form) · sensor fusion

The tool that makes the proofs clean: the **characteristic function**.

THE CHARACTERISTIC FUNCTION

Why. Showing that $Ax + b + \varepsilon$ is *Gaussian* with densities needs a change of variables (breaks when A is not square/invertible) and a convolution. The characteristic function avoids both.

Definition (Characteristic function). For a random vector $X \in \mathbb{R}^n$,

$$\varphi_X(\xi) = \mathbb{E}[e^{i\xi^\top X}], \quad \xi \in \mathbb{R}^n,$$

i.e. the Fourier transform of its distribution.

What ξ is. Not random — a fixed "frequency" vector, the free argument. $\xi^\top X$ projects X onto direction ξ ; knowing φ_X along every direction and scale pins down the distribution.

Why it always exists. $|e^{i\theta}| = 1$, so $\mathbb{E}[|e^{i\xi^\top X}|] = 1 < \infty$ for every X . The moment generating function (MGF) $\mathbb{E}[e^{\xi^\top X}]$ can be infinite (e.g. Cauchy distribution).

THREE PROPERTIES

1. **Uniqueness.** Same characteristic function $\Rightarrow$ same distribution. To show y is Gaussian, show φ_y has the Gaussian form.
2. **Sums $\rightarrow$ products.** If $X \perp Z$: $\varphi_{X+Z}(\xi) = \varphi_X(\xi) \varphi_Z(\xi)$. Convolution becomes multiplication.
3. **Linear maps.** $\varphi_{AX+b}(\xi) = e^{i\xi^\top b} \varphi_X(A^\top \xi)$, for any A , square or not.

The Gaussian case. If $X \sim \mathcal{N}(\mu, \Sigma)$:

$$\varphi_X(\xi) = \exp(i\xi^\top \mu - \frac{1}{2}\xi^\top \Sigma \xi)$$

DEGENERATE GAUSSIANS

Definition. Lecture 4 defined $\mathcal{N}(\mu, \Sigma)$ by its density, which needs $\Sigma \succ 0$. The characteristic function makes sense for any $\Sigma \succeq 0$ and serves as the **general definition**. Singular Σ : no density; all mass on an affine subspace.

Where they arise.

- $y = Ax$ with A having more rows than its rank $\rightarrow A\Sigma A^\top$ singular.
- The joint of (x_{t-1}, x_t) with $Q = 0$: $x_t = Ax_{t-1} + Ba_{t-1}$ exactly.

Closure theorems are proved with characteristic functions, so they hold for every $\Sigma \succeq 0$ — and so does the Kalman filter.

WHAT A DEGENERATE GAUSSIAN KNOWS EXACTLY

The variance along a direction. For any fixed $u \in \mathbb{R}^n$, the scalar $u^\top x$ has mean $u^\top \mu$, so

$$\text{Var}(u^\top x) = \mathbb{E}[(u^\top x - u^\top \mu)^2] = \mathbb{E}[u^\top (x - \mu)(x - \mu)^\top u] = u^\top \mathbb{E}[(x - \mu)(x - \mu)^\top] u = u^\top \Sigma u \geq 0.$$

This is the Lec 4 proof that covariance is PSD.

A null direction u , with $\Sigma u = 0$, therefore gives $\text{Var}(u^\top x) = 0$:

$$u^\top x = u^\top \mu \quad \text{with probability 1.}$$

So the combination $u^\top x$ is **known exactly**, even though x itself is random.

- Each null direction is one exact **linear equation** on the state. With k independent null directions, x lives on the affine subspace through μ of dimension $n - k = \text{rank } \Sigma$.
- A **single coordinate** x_i is known exactly only if e_i is a null direction: the i -th column of Σ is zero. For a covariance this is the same as $\Sigma_{ii} = 0$, since $|\Sigma_{ij}| \leq \sqrt{\Sigma_{ii}\Sigma_{jj}}$. In general, no coordinate is certain — only a combination is.

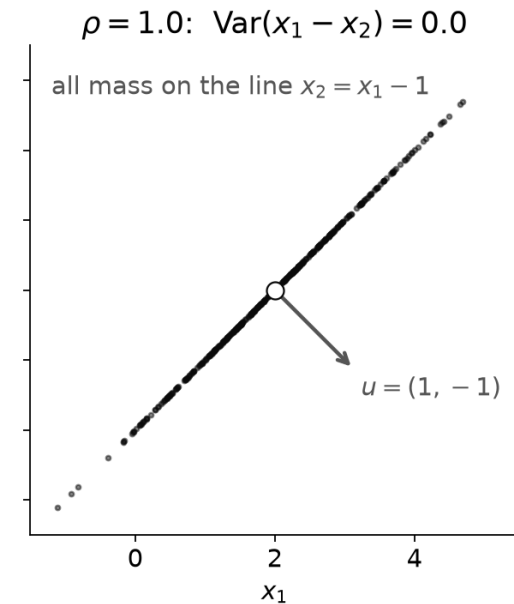
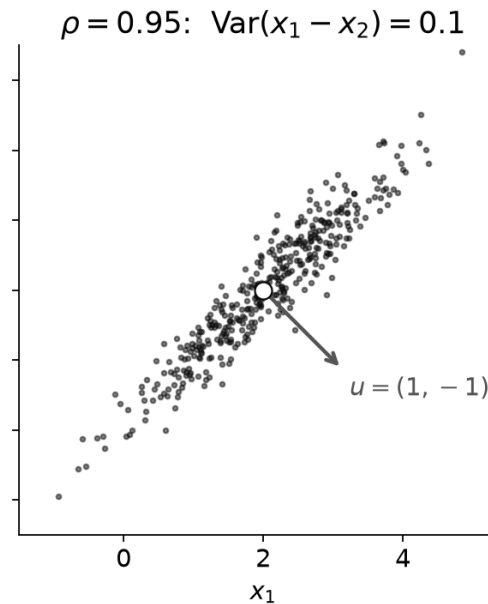
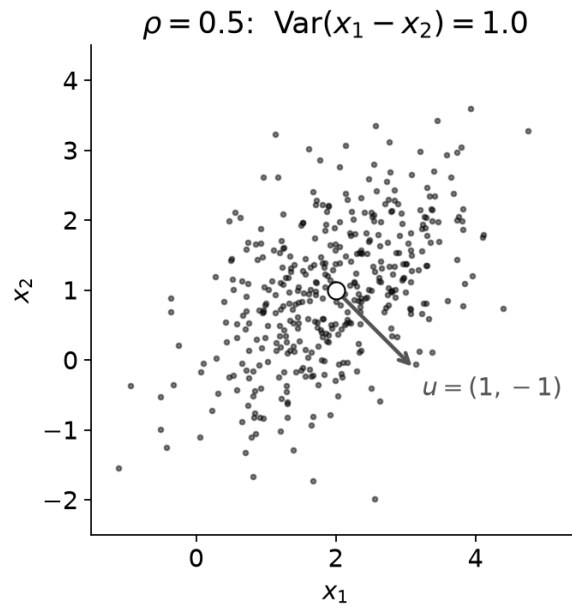
EXAMPLE – THE DIFFERENCE IS KNOWN, NEITHER VALUE IS

$$\Sigma = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \text{eigenvalues } 2 \text{ along } (1, 1), \quad 0 \text{ along } u = (1, -1).$$

- **Certain:** $\text{Var}(x_1 - x_2) = u^\top \Sigma u = \Sigma_{11} + \Sigma_{22} - 2\Sigma_{12} = 1 + 1 - 2 = 0$, so $x_1 - x_2 = \mu_1 - \mu_2$.
- **Uncertain:** $\text{Var}(x_1) = \text{Var}(x_2) = 1$. Where the pair sits along $(1, 1)$ is unknown.
- **One noise source drives both:** $x = \mu + z(1, 1)$, $z \sim \mathcal{N}(0, 1)$, has exactly this Σ .
- **Learning one coordinate fixes the other:** $x_2 = x_1 - (\mu_1 - \mu_2)$.

FROM AN ELLIPSE TO A LINE

$$\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \mu = (2, 1): \text{Var}(x_1 - x_2) = 2 - 2\rho.$$



Nearly degenerate is the common case. As $\rho \rightarrow 1$ the ellipse thins into the line. Tiny process noise or a very precise sensor gives an almost-singular Σ — where the information form breaks numerically and the Joseph form (§4) helps.

DEGENERATE GAUSSIANS IN THE FILTER

Picture: two robots on a rigid bar of length 1. The difference of their positions is known exactly; where the bar is, is not.

The $Q = 0$ joint is the same pattern. With no process noise,

$$x_t - Ax_{t-1} = Ba_{t-1} \quad \text{exactly,}$$

so the joint of (x_{t-1}, x_t) has null directions $(-A^\top v, v)$ for every $v \in \mathbb{R}^n$. The *relation* between consecutive states is certain; neither state is.

DEGENERATE GAUSSIANS ENCODE LINEAR CONSTRAINTS

A degenerate Gaussian encodes exact **linear holonomic constraints** on the state.

Curved constraints (pendulum rod, $x^2 + y^2 = \ell^2$) can only be matched locally:

- a Gaussian over (x, y) leaks mass off the circle (wrong rod lengths look plausible);
- its mean falls *inside* the circle: $\|\mathbb{E}[(x, y)]\| < \ell$.

Fix: choose coordinates. In θ every value is a valid configuration (why Lec 2 prefers generalised coordinates). But:

- angles wrap at $\pm\pi \rightarrow$ a Gaussian on θ only works for small uncertainty;
- 3D orientation has no singularity-free minimal coordinates $\rightarrow$ attitude filters estimate a small error in local coordinates.

Nonholonomic constraints (unicycle "no sideways slip") restrict velocities, not reachable states. They belong in $\mathcal{f}$, not in Σ .

WHY THE FILTER NEEDS DEGENERATE GAUSSIANS

Gaussians *with* a density are **not closed** under the filter's operations.

- **Start degenerate.** Robot at a known dock: $\Sigma_0 = 0$. A known parameter carried in the state.
- **Become degenerate.** A noise-free sensor channel makes the measured direction exactly known.
- **Recover.** Position-velocity with noise on velocity only:

$$A = \begin{pmatrix} 1 & \Delta t \\ 0 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & 0 \\ 0 & q \end{pmatrix} : \quad \bar{\Sigma}_1 = Q \text{ (singular)}, \quad \bar{\Sigma}_2 = q \begin{pmatrix} \Delta t^2 & \Delta t \\ \Delta t & 2 \end{pmatrix}, \quad \det = q^2 \Delta t^2 > 0.$$

Velocity noise leaks into position through A .

Degenerate members are the points where the belief passes between "known exactly" and "uncertain".

LINEAR TRANSFORMATION

Theorem (Linear transformation). Let $x \sim \mathcal{N}(\mu, \Sigma)$ in $\mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ fixed, $\varepsilon \sim \mathcal{N}(0, R)$ independent of x . Then

$$y = Ax + b + \varepsilon \sim \mathcal{N}(A\mu + b, A\Sigma A^\top + R).$$

A need not be square or invertible.

Proof. Properties 2 and 3:

$$\varphi_y(\xi) = e^{i\xi^\top b} \varphi_x(A^\top \xi) \varphi_\varepsilon(\xi) = e^{i\xi^\top (A\mu + b) - \frac{1}{2}\xi^\top (A\Sigma A^\top + R)\xi}.$$

By uniqueness (property 1), $y \sim \mathcal{N}(A\mu + b, A\Sigma A^\top + R)$. $\square$

NON-CENTRED NOISE = BIAS

If $\varepsilon \sim \mathcal{N}(c, R)$, write $\varepsilon = c + \varepsilon'$:

$$y \sim \mathcal{N}(A\mu + b + c, A\Sigma A^\top + R).$$

The offset only shifts the mean and can be absorbed into b — zero-mean noise loses no generality.

In a filter, a nonzero noise mean is a **bias**:

- **Known:** subtract it — add c to Ba_{t-1} in predict, or use $o_t - c$ in update.
- **Unknown and ignored:** estimates systematically offset; innovations have nonzero mean.
- **Standard fix:** state augmentation — add the bias to the state and estimate it.

MARGINALISATION AND CONDITIONING

For a jointly Gaussian pair

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}\right) :$$

Theorem (Marginalisation). $x_1 \sim \mathcal{N}(\mu_1, \Sigma_{11})$.

Proof. Linear transformation $x_1 = [I \ 0] x$ with $\varepsilon = 0$. $\square$

Theorem (Conditioning). For $\Sigma_{22} \succ 0$,

$$x_1 \mid x_2 = z \sim \mathcal{N}\left(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(z - \mu_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\right).$$

$\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$ is the **Schur complement** of Σ_{22} .

PROOF OF CONDITIONING – DECORRELATION TRICK (1-3)

Idea: split x_1 into a linear function of x_2 plus a remainder e **independent** of x_2 . Let $M = \Sigma_{12}\Sigma_{22}^{-1}$.

1. **Define the remainder** $e = x_1 - Mx_2$. Then

$$\text{Cov}(e, x_2) = \Sigma_{12} - M\Sigma_{22} = 0.$$

M is the only matrix that does this.

2. **Jointly Gaussian.** $\begin{pmatrix} e \\ x_2 \end{pmatrix} = \begin{pmatrix} I & -M \\ 0 & I \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ – a linear map of a Gaussian.

3. **Uncorrelated + jointly Gaussian $\Rightarrow$ independent.** Block-diagonal covariance makes the characteristic function factorise:

$$\varphi_{(e,x_2)}(\xi_1, \xi_2) = \varphi_e(\xi_1) \varphi_{x_2}(\xi_2) \Rightarrow e \perp x_2.$$

Joint Gaussianity is essential: in general, zero correlation $\nRightarrow$ independence.

PROOF OF CONDITIONING (4-5)

4. **Condition.** $x_1 = e + Mx_2$. Given $x_2 = z$:

- $Mx_2 = Mz$ is a fixed vector;
- by independence, $e \mid x_2 = z \sim \mathcal{N}(m_e, \Sigma_e)$ – unchanged.

So $x_1 \mid x_2 = z \sim \mathcal{N}(m_e + Mz, \Sigma_e)$.

5. **Compute m_e and Σ_e .** Mean: $m_e + Mz = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(z - \mu_2)$. Covariance, reusing step 1:

$$\Sigma_e = \text{Cov}(e, x_1 - Mx_2) = \text{Cov}(e, x_1) - \underbrace{\text{Cov}(e, x_2) M^\top}_{=0} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}. \quad \blacksquare$$

Remark (M is the regression coefficient). $M = \Sigma_{12}\Sigma_{22}^{-1}$ also minimises $\mathbb{E}\|x_1 - Mx_2 - c\|^2$: the optimality condition is $\text{Cov}(e, x_2) = 0$ – the **orthogonality principle**. *For Gaussians, the best linear predictor and the conditional mean coincide.*

CONDITIONING IS BAYESIAN UPDATING

Let $x_1 = x$ (hidden state), $x_2 = o$ (observation), z = the actual reading. Then $p(x \mid o = z)$ is the **posterior**: the conditioning theorem *is* Bayes' rule for Gaussians, in closed form.

SYMBOL	MEANING IN ESTIMATION
μ_1, Σ_{11}	prior belief: best guess and its uncertainty
μ_2	predicted observation
Σ_{22}	uncertainty of that prediction: prior seen through the sensor + sensor noise
Σ_{12}	how much a reading says about the state
$z - \mu_2$	the surprise
$\Sigma_{12}\Sigma_{22}^{-1}$	the gain : how far to move per unit of surprise

READING THE FORMULA

- **Posterior mean = prior mean + gain $\times$ surprise.** If $z = \mu_2$ the estimate does not move. Large gain when the observation is strongly tied to the state (large Σ_{12}) and reliable (small Σ_{22}).
- **Posterior covariance = prior covariance – what the observation explains.** $\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} \succeq 0$: **observing never increases uncertainty.**
- **The posterior covariance does not depend on z .** The shrinkage is fixed by *what* the sensor measures and how noisy it is — not by *what it happened to read*.

Corollary (Gaussian beliefs stay Gaussian). If b_0 is Gaussian and f, h are linear with additive Gaussian noise, every $\bar{b}_t$ and b_t is Gaussian. Predict uses the linear transformation theorem; update uses conditioning.

This is a theorem, not a modelling convenience — the reason the Kalman filter is exact and closed-form.

CONJUGATE PRIORS

Why. "Prior $\times$ likelihood, then normalise" generally produces an unnamed function whose shape grows more complicated every step. Conjugacy stops this: the filter runs indefinitely with **fixed memory and no integrals**.

Definition (Conjugate prior). A family $\mathcal{P}$ is **conjugate** to a likelihood $p(o \mid x)$ if, for every prior in $\mathcal{P}$, the posterior is also in $\mathcal{P}$.

Lemma (Completing the square). Let $\Lambda \succ 0$ symmetric and $\eta \in \mathbb{R}^n$. If

$$p(x) \propto \exp\left(-\frac{1}{2}x^\top \Lambda x + \eta^\top x\right),$$

then $p = \mathcal{N}(\Lambda^{-1}\eta, \Lambda^{-1})$. For a Gaussian, Λ is the **precision** Σ^{-1} and η the **information vector** $\Sigma^{-1}\mu$.

Proof. Expand $-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu) = -\frac{1}{2}x^\top \Lambda x + \eta^\top x + \text{const}$. Proportional densities are equal. $\square$

GAUSSIAN-GAUSSIAN CONJUGACY

Proposition. Prior $x \sim \mathcal{N}(\mu_0, \Sigma_0)$, $\Sigma_0 \succ 0$. Linear Gaussian sensor $o = Hx + v$, $v \sim \mathcal{N}(0, R)$, $R \succ 0$, $v \perp x$. Then $p(x | o) = \mathcal{N}(\mu_1, \Sigma_1)$ with two equivalent forms:

Information form ($\Lambda = \Sigma^{-1}$, $\eta = \Sigma^{-1}\mu$):

$$\Lambda_1 = \Lambda_0 + H^\top R^{-1} H, \quad \eta_1 = \eta_0 + H^\top R^{-1} o.$$

Gain form ($S = H\Sigma_0 H^\top + R$, $K = \Sigma_0 H^\top S^{-1}$):

$$\mu_1 = \mu_0 + K(o - H\mu_0), \quad \Sigma_1 = (I - KH)\Sigma_0.$$

Why η rather than μ ? In natural parameters the update is **additive**: the observation contributes $H^\top R^{-1} H$ to the precision and $H^\top R^{-1} o$ to the information vector. Recovering $\mu_1 = \Lambda_1^{-1} \eta_1$ needs an inverse — compute only when needed.

PROOF – INFORMATION FORM

By Bayes' rule, $\log p(x \mid o) = \log p(o \mid x) + \log p(x) + \text{const}$:

$$\log p(x \mid o) = -\frac{1}{2}(o - Hx)^\top R^{-1}(o - Hx) - \frac{1}{2}(x - \mu_0)^\top \Sigma_0^{-1}(x - \mu_0) + \text{const}.$$

Expand both quadratics, keep terms in x :

$$\log p(x \mid o) = -\frac{1}{2} x^\top \underbrace{(\Sigma_0^{-1} + H^\top R^{-1} H)}_{\Lambda_1} x + x^\top \underbrace{(\Sigma_0^{-1} \mu_0 + H^\top R^{-1} o)}_{\eta_1} + \text{const}.$$

$\Lambda_1 \succ 0$ (PD plus PSD), so the lemma gives $p(x \mid o) = \mathcal{N}(\Lambda_1^{-1} \eta_1, \Lambda_1^{-1})$. $\square$

PROOF – FROM INFORMATION FORM TO GAIN FORM

Woodbury identity. $\Lambda_1^{-1} = (\Sigma_0^{-1} + H^\top R^{-1} H)^{-1} = \Sigma_0 - \Sigma_0 H^\top S^{-1} H \Sigma_0$.

Check: multiply out $\Lambda_1 W$ for W the right-hand side:

$$\Lambda_1 W = I + H^\top \left[R^{-1} - S^{-1} - R^{-1} H \Sigma_0 H^\top S^{-1} \right] H \Sigma_0 = I + H^\top R^{-1} (S - R - H \Sigma_0 H^\top) S^{-1} H \Sigma_0 = I.$$

Gain identity. $\Sigma_1 H^\top R^{-1} = K$:

$$\Sigma_1 H^\top R^{-1} = \Sigma_0 H^\top \left[I - S^{-1} H \Sigma_0 H^\top \right] R^{-1} = \Sigma_0 H^\top S^{-1} \underbrace{\left[S - H \Sigma_0 H^\top \right]}_R R^{-1} = K.$$

Covariance. $\Sigma_1 = \Sigma_0 - K H \Sigma_0 = (I - K H) \Sigma_0$.

Mean. $\mu_1 = \Sigma_1 \Sigma_0^{-1} \mu_0 + \Sigma_1 H^\top R^{-1} o = (\mu_0 - K H \mu_0) + K o = \mu_0 + K(o - H \mu_0)$. ■

TWO FORMS, TWO DOMAINS

Check against §3.3. The joint of (x, o) has $\Sigma_{12} = \Sigma_0 H^\top, \Sigma_{22} = S \rightarrow$ conditioning gives the gain form directly.

	INFORMATION FORM	GAIN FORM
Inverts	Σ_0, R (both must be PD)	only $S (d \times d)$
Degenerate prior $\Sigma_0 = 0$	$\times$ undefined	$\checkmark$
Uninformative prior $\Lambda_0 = 0$	$\checkmark$	$\times$ cannot express
Story	precisions add	move toward the measurement by a fraction K of the surprise
Natural for	sensor fusion (Lec 7, §5)	the Kalman filter (§4)

Lecture 4 (§5.2) derived the same posterior mean as Tikhonov-regularised least squares.

Conjugacy alone covers only the update. The Gaussian matters because it is **also closed under predict**.

§4 · THE KALMAN FILTER

THE BAYES FILTER, LINEAR AND GAUSSIAN



THE LINEAR GAUSSIAN MODEL

$$x_t = Ax_{t-1} + Ba_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, Q)$$

$$o_t = Hx_t + v_t, \quad v_t \sim \mathcal{N}(0, R)$$

$A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $H \in \mathbb{R}^{d \times n}$, $Q \succeq 0$, $R \succ 0$. Noises and $x_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$ mutually independent and independent over time.

Where the model comes from.

- **Dynamics:** A, B = discretised linearisation (Jacobians, Lec 3 §5; ZOH, Lec 3 §6.2). Q models everything the linear model leaves out.
- **Observation:** H selects/combines measured components, e.g. $H = [I \ 0]$ = "position measured, velocity not". R from the datasheet or calibration.

$Q \succeq 0$ allows noise on some directions only; $R \succ 0$ keeps S_t invertible.

PREDICT STEP

$$\bar{\mu}_t = A\mu_{t-1} + Ba_{t-1}, \quad \bar{\Sigma}_t = A\Sigma_{t-1}A^\top + Q$$

Derivation. Linear transformation theorem with random vector $x_{t-1} \sim \mathcal{N}(\mu_{t-1}, \Sigma_{t-1})$, matrix A , fixed vector $b = Ba_{t-1}$ (action known), noise $\varepsilon = w_t$.

Interpretation. The mean follows the noise-free dynamics. $A\Sigma A^\top$ stretches and rotates the ellipsoid (shrinks along stable, grows along unstable directions); Q adds fresh uncertainty. Without observations, dead reckoning **drifts**.

Prediction comes for free. k predict steps without updates:

$$\mu_{t+k|t} = A^k \mu_t + \sum_{j=0}^{k-1} A^{k-1-j} B a_{t+j}, \quad \Sigma_{t+k|t} = A^k \Sigma_t (A^k)^\top + \sum_{j=0}^{k-1} A^j Q (A^j)^\top.$$

The cone of uncertainty for collision checking (Lec 9) and MPC predictions (Lec 11).

UPDATE STEP – THE JOINT

Under $\bar{b}_t$, $x_t \sim \mathcal{N}(\bar{\mu}_t, \bar{\Sigma}_t)$ and $o_t = Hx_t + v_t$. Stack:

$$\begin{pmatrix} x_t \\ o_t \end{pmatrix} = \begin{pmatrix} I \\ H \end{pmatrix} x_t + \begin{pmatrix} 0 \\ v_t \end{pmatrix}$$

The noise is degenerate (§3.2).

Linear transformation theorem:

$$\begin{pmatrix} x_t \\ o_t \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \bar{\mu}_t \\ H\bar{\mu}_t \end{pmatrix}, \begin{pmatrix} \bar{\Sigma}_t & \bar{\Sigma}_t H^\top \\ H\bar{\Sigma}_t & S_t \end{pmatrix}\right), \quad S_t = H\bar{\Sigma}_t H^\top + R.$$

- $H\bar{\mu}_t$: the **predicted reading**.
- $\text{Cov}(x_t, o_t) = \bar{\Sigma}_t H^\top$, since $v_t \perp x_t$.
- $\text{Cov}(o_t) = S_t$: prior uncertainty seen through the sensor plus sensor noise.

UPDATE STEP – CONDITIONING

Conditioning theorem with $x_1 = x_t, x_2 = o_t$:

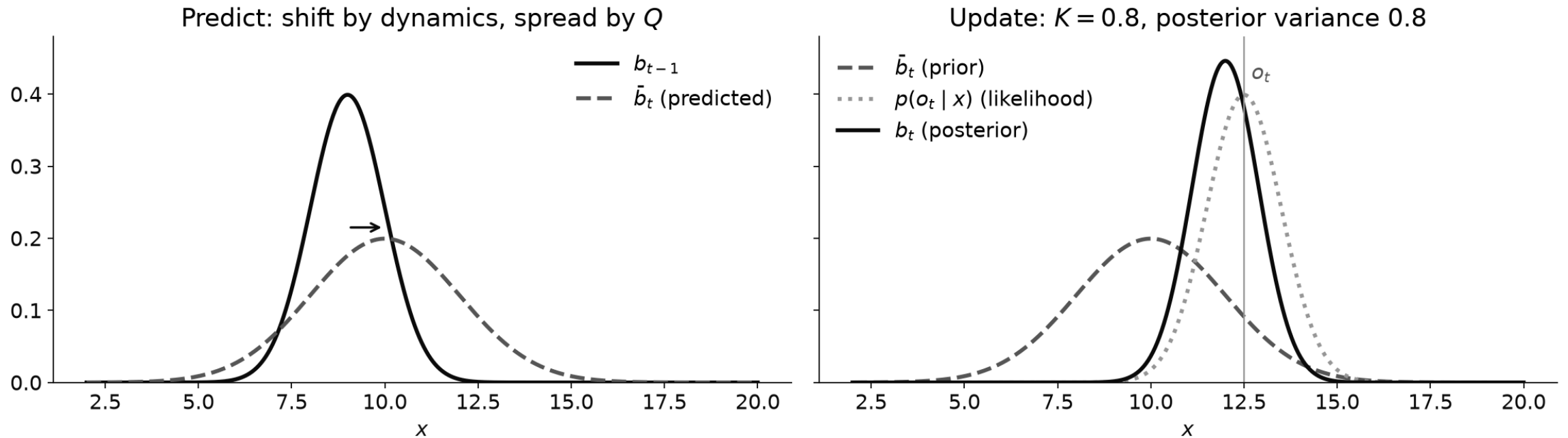
$$S_t = H\bar{\Sigma}_t H^\top + R, \quad K_t = \bar{\Sigma}_t H^\top S_t^{-1}, \quad \mu_t = \bar{\mu}_t + K_t(o_t - H\bar{\mu}_t), \quad \Sigma_t = (I - K_t H) \bar{\Sigma}_t$$

Equivalently: Gaussian–Gaussian conjugacy (§3.4) with prior $\bar{b}_t$.

Vocabulary.

- **Innovation** $\nu_t = o_t - H\bar{\mu}_t$: the part of the observation the prediction did not anticipate — the only new information o_t brings.
- **Innovation covariance** S_t : how large the innovation should be.
- **Kalman gain** $K_t \in \mathbb{R}^{n \times d}$: maps innovations in observation space to corrections in state space.

PREDICT AND UPDATE, PICTURED



Scalar: $\bar{\sigma}^2 = 4, r = 1 \rightarrow K = 0.8$. The posterior is narrower than both prior and likelihood, and sits closer to the more reliable source.

INTERPRETING THE KALMAN GAIN

Scalar case ($H = 1$, prior variance $\bar{\sigma}^2$, sensor variance r):

$$K = \frac{\bar{\sigma}^2}{\bar{\sigma}^2 + r} \in (0, 1), \quad \mu = (1 - K) \bar{\mu} + K o, \quad \sigma^2 = (1 - K) \bar{\sigma}^2.$$

A convex combination, each source weighted by the *other's* variance: **the more reliable source gets more weight.**

Limiting cases.

- $R \rightarrow 0$ (perfect sensor), H invertible: $K_t \rightarrow H^{-1}$, state read off the observation. Non-invertible H : only the seen components are corrected.
- $\bar{\Sigma}_t \rightarrow 0$ (perfect prediction): $K_t \rightarrow 0$, observation ignored.

Covariance only shrinks. $\bar{\Sigma}_t - \Sigma_t = \bar{\Sigma}_t H^\top S_t^{-1} H \bar{\Sigma}_t \succeq 0$, concentrated along directions the sensor sees.

Hidden states are corrected through correlation. Predict creates position–velocity correlation in $\bar{\Sigma}_t \rightarrow$ the gain's velocity rows are nonzero $\rightarrow$ a position innovation corrects velocity.

THE INNOVATION IS ZERO. DID THE UPDATE DO NOTHING?



THE ALGORITHM

Algorithm (Kalman filter).

Input: $b_0 = \mathcal{N}(\mu_0, \Sigma_0)$; model A, B, H, Q, R ; stream $(a_0, o_1), (a_1, o_2), \dots$

For $t = 1, 2, \dots$

1. **Predict:** $\bar{\mu}_t = A\mu_{t-1} + Ba_{t-1}, \quad \bar{\Sigma}_t = A\Sigma_{t-1}A^\top + Q$
2. **Update:** $S_t = H\bar{\Sigma}_tH^\top + R, \quad K_t = \bar{\Sigma}_tH^\top S_t^{-1}, \quad \mu_t = \bar{\mu}_t + K_t(o_t - H\bar{\mu}_t), \quad \Sigma_t = (I - K_tH)\bar{\Sigma}_t$

Output: $b_t = \mathcal{N}(\mu_t, \Sigma_t)$.

Cost per step: $O(n^3 + d^3)$ time, $O(n^2 + d^2)$ memory, **independent of t** . n^3 from $A\Sigma A^\top$, d^3 from inverting S_t . Only the current (μ_t, Σ_t) is stored.

Zero innovation: the mean does not move, but the covariance still shrinks — seeing the expected value is still information.

WHAT'S NEXT

Lecture 7 — State Estimation (III)

- §4 (cont.) Why the Kalman filter is optimal · the covariance ignores the data · observability and the long run
- §5 Sensor fusion — precisions add
- §6 EKF / UKF and the particle filter
- §7 The belief as a sufficient statistic: estimator + controller